TOC LAB

WAP in C to count number of 1’s and 0’s in binary string

```
anil@linux:~/Desktop/Code$ gcc count_0sand1s.c -o count_0sand1s
anil@linux:~/Desktop/Code$ ./count_0sand1s
Enter a binary string in 0's and 1's :101010
Number of 0s in String :3
Number of 1s in String :3
anil@linux:~/Desktop/Code$
```

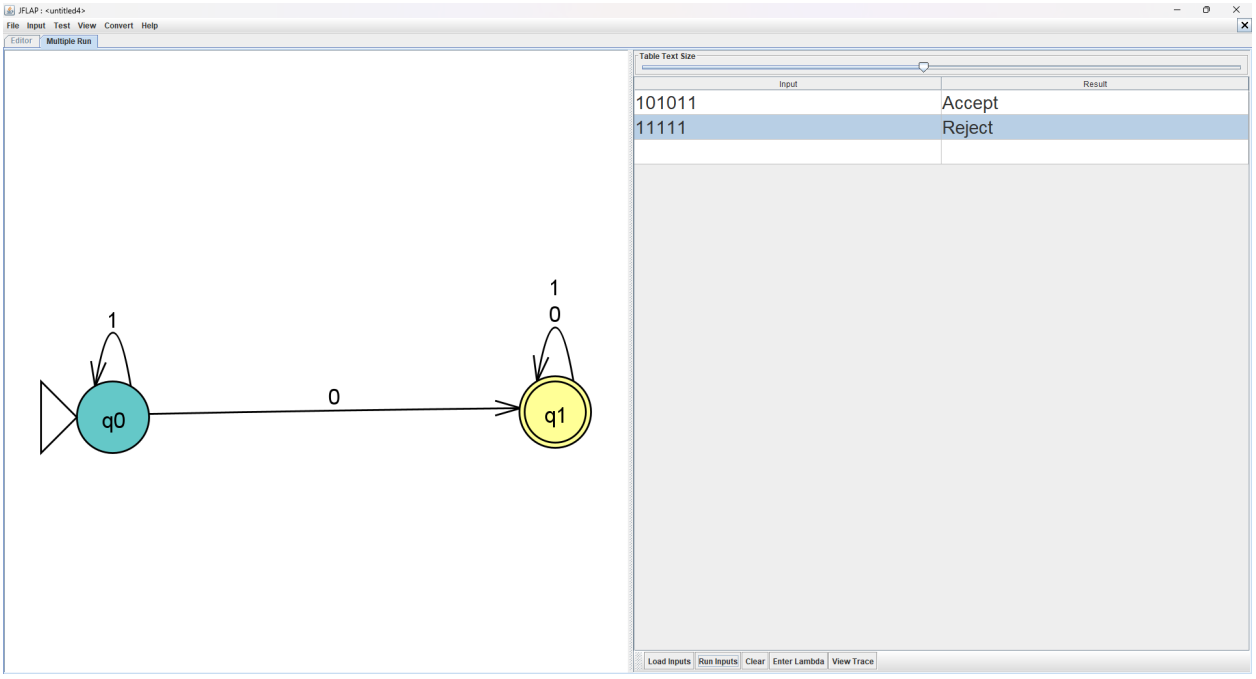
WAP in C to determine that the 1’s and 0’s are even or odd in a binary string

```
anil@linux:~/Desktop/Code$ gcc Odd_Even_0sand1s.c -o Odd_Even_0sand1s
anil@linux:~/Desktop/Code$ ./Odd_Even_0sand1s
Enter a binary string in 0's and 1's :0000011001101101
Odd number of 0's .
Odd number of 1's .
anil@linux:~/Desktop/Code$
```

WAP ot check if entered string is Binary or not

```c
#include <stdio.h>
#include <string.h>

int main() {
    char binary[100];
    int i = 0, valid = 1;

    printf("Enter the Binary String: ");
    scanf("%s", binary);

    while (binary[i] != '\0') {
        if (binary[i] != '0' && binary[i] != '1') {
            valid = 0;   // mark as invalid
            break;       // stop checking further
        }
        i++;
    }

    if (valid)
        printf("String is Accepted.\n");
    else
        printf("String is Not Accepted.\n");

    return 0;
}
```

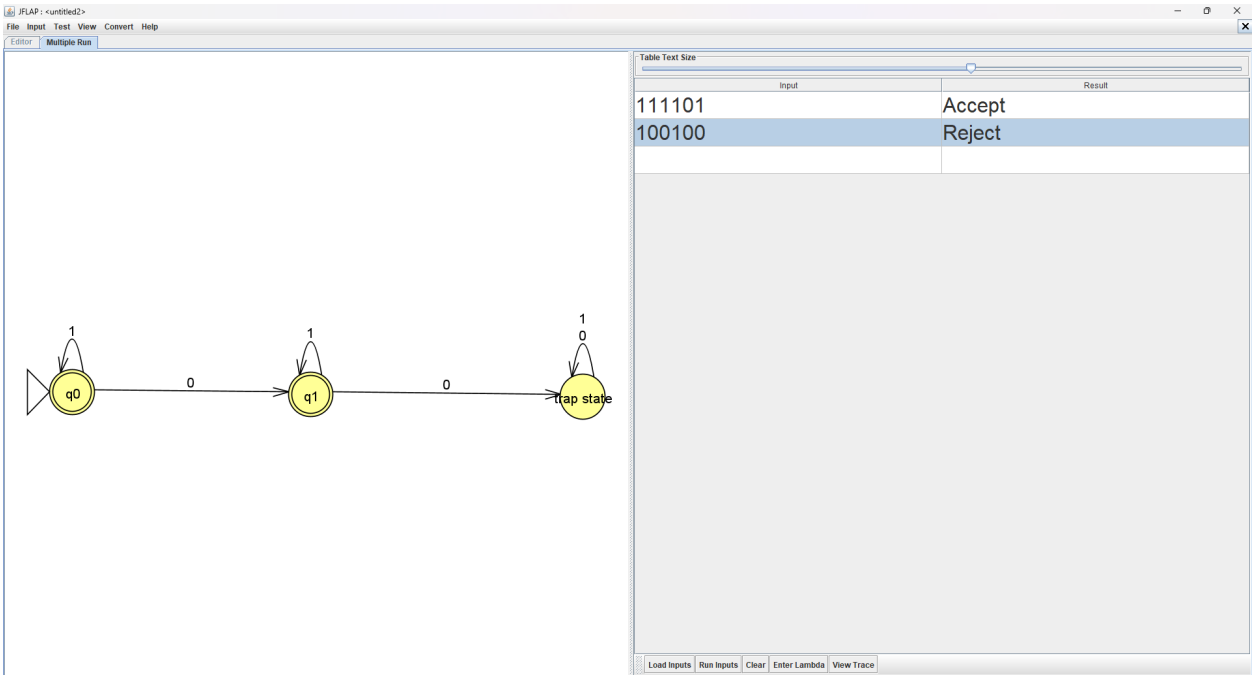
WAP to create a DFA machine which accepts the string having atleast one 0 in it.



Expected Output :-

```
anil:ForTOC/ $ gcc Atleast_one_zero.c -o Atleast_one_zero                    [10:27:20]
anil:ForTOC/ $ ./Atleast_one_zero                                            [10:27:55]
Enter your Binary String (0/1):1010111111
Accepted
anil:ForTOC/ $ ./Atleast_one_zero                                            [10:28:19]
Enter your Binary String (0/1):1111111111
Rejected
anil:ForTOC/ $                                                               [10:28:25]
```

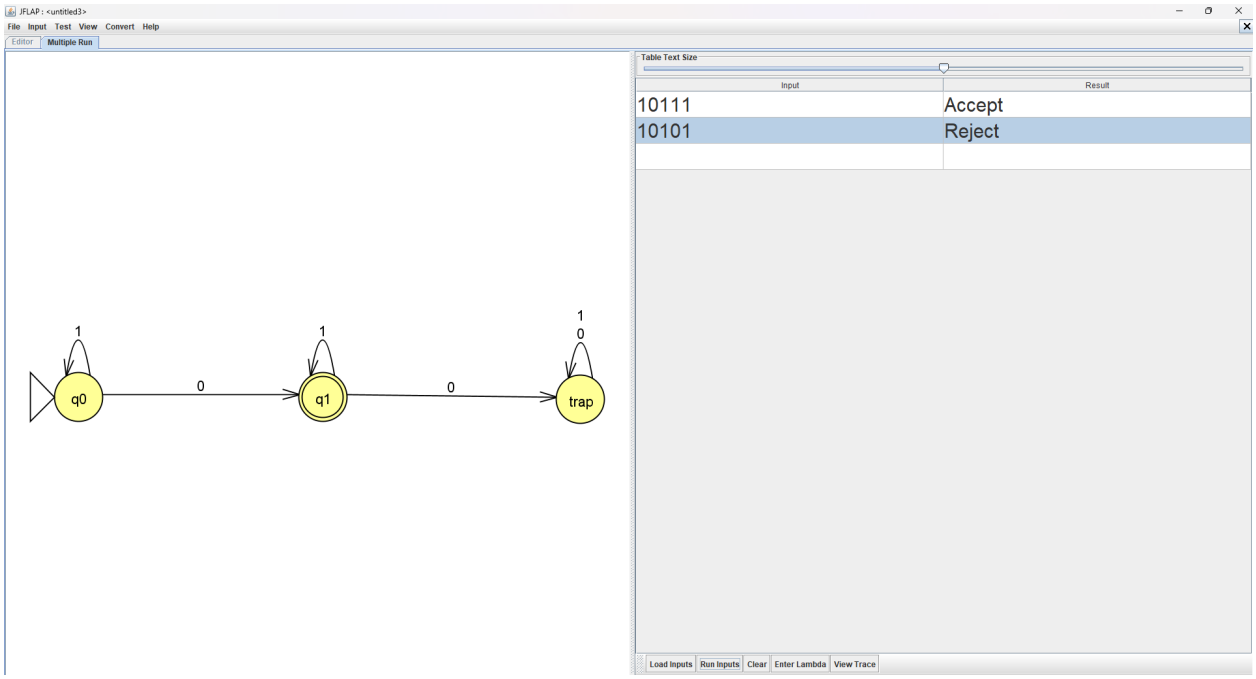
WAP to create a DFA machine which accepts the string having atmost one 0 in it.



Expected Output :-

```
anil:ForTOC/ $ gcc Atmost_one_zero_c.c -o Atmost_one_zero_c                    [11:56:28]
anil:ForTOC/ $ ./Atmost_one_zero_c                                             [11:57:27]
Enter a binary string: 1010111111
Rejected : String has more than one '0'
anil:ForTOC/ $ ./Atmost_one_zero_c                                             [11:57:38]
Enter a binary string: 11111111110
Accepted : String has at most one '0'
anil:ForTOC/ $                                                                 [11:57:46]
```

WAP to create a DFA machine which accepts the string having exactly one 0 in it.



Expected Output :-

```
anil:ForTOC/ $ ./Exact_one_zero
                        [13:04:05]
Enter a binary string: 00110011
Rejected: String does not have exactly one '0'.
anil:ForTOC/ $ ./Exact_one_zero
                        [13:04:17]
Enter a binary string: 01111111111111
Accepted: String has exactly one '0'.
```